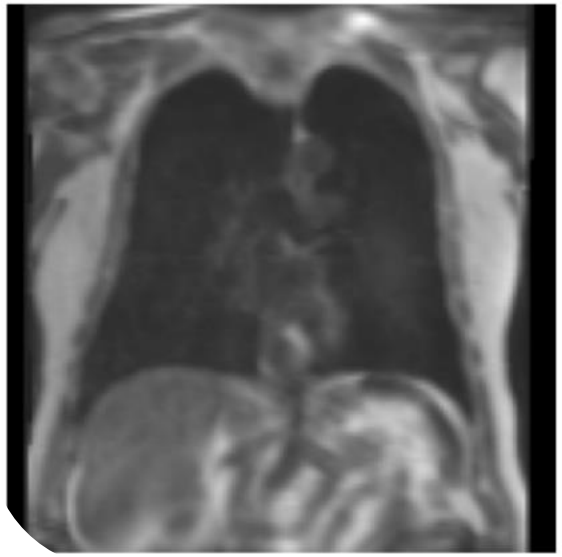
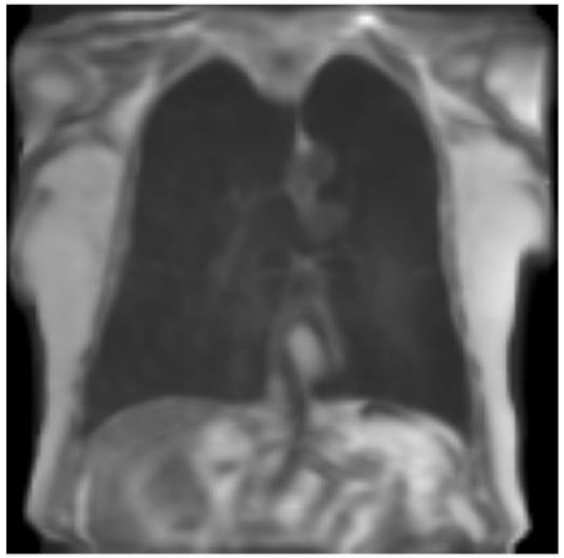
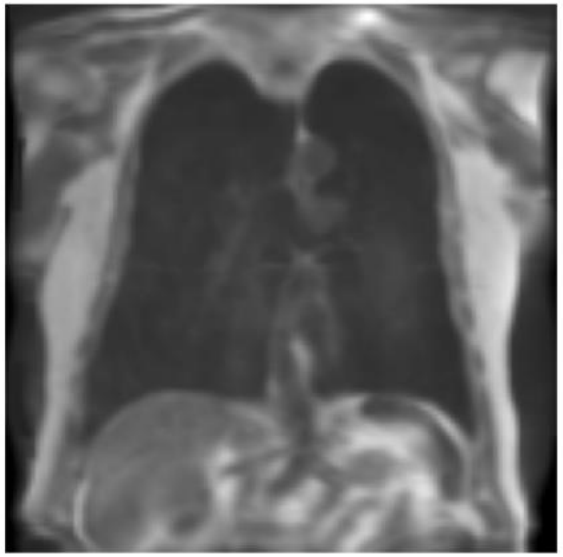
Registration of Lung MRI with Machine Learning Methods

Chieh-Han, Lee

25th October, 2024

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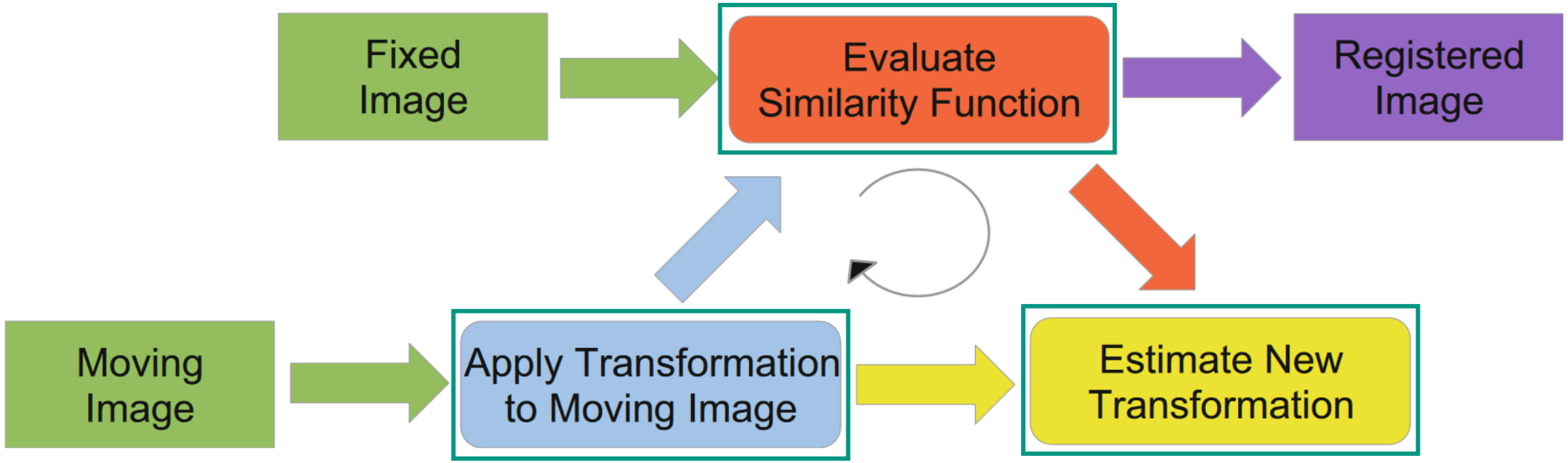
Motivation

- A crucial preprocessing step to medical image application and tasks
  - Standard basis for medical image analysis
- Challenges in Lung MRI registration due to lungs' passive nature
  - Easily affected by respiration and heartbeat
- Improvement in speed
  - Deep Learning Methods provides faster registration with comparable results
- Foundation for future research
  - More research can be done with out preliminary model

State of Research – Medical Image Registration

- Aligns two or more related images of different modalities and/or time points to produce a comprehensive single image.



Min Chen, 2023

State of Research

- **Traditional Methods**
  - Advanced Normalization Tools (ANTs)
  - Elastix
- **Deep Learning Methods**
  - Convolution Models
  - Transformer Models

State of Research – Convolution Models

- VoxelMorph and CycleMorph
- Utilizes CNNs
- Local feature extraction
- Hierarchical features
- Fixed perceptive field

State of Research – Transformer Models

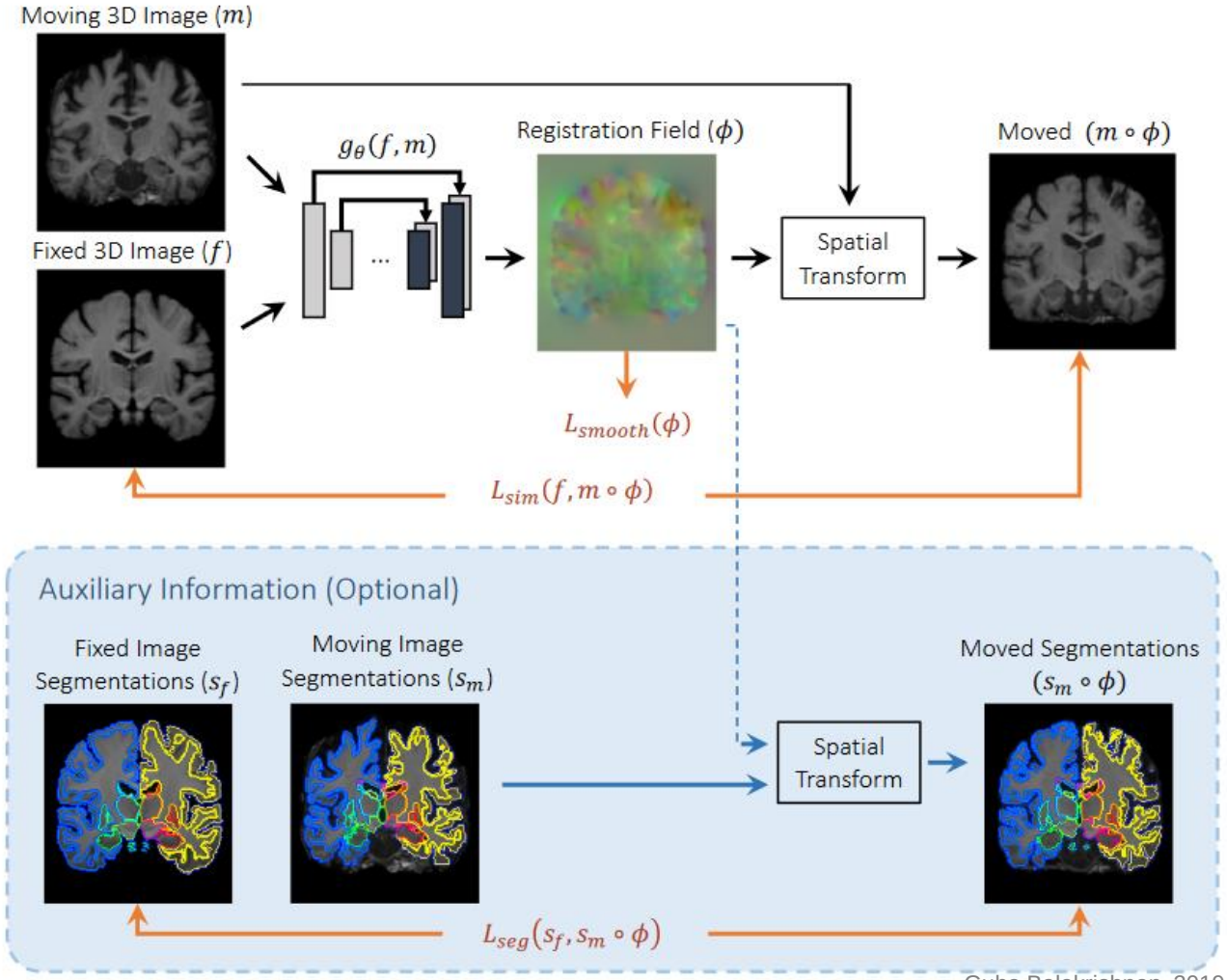
- ViT-V-Net & TransMorph
- Utilizes Transformer
- Attention Mechanism: Enables the model to focus on each input data relative to each other (e.g., pixels in an image or words in a sentence).
- Adjusts the weight of each input based on contextual information.
- Improvement of the network’s Perception Field

State of Research – Network Decision

- VoxelMorph
  - Fundamental image registration network
- CycleMorph
  - Parallel, cycle consistent CNN model
- ViT-V-Net
  - Vision Transformer implementation at the lowest level
- TransMorph
  - Increase the use of Transformer for further improvements

State of Research – VoxelMorph

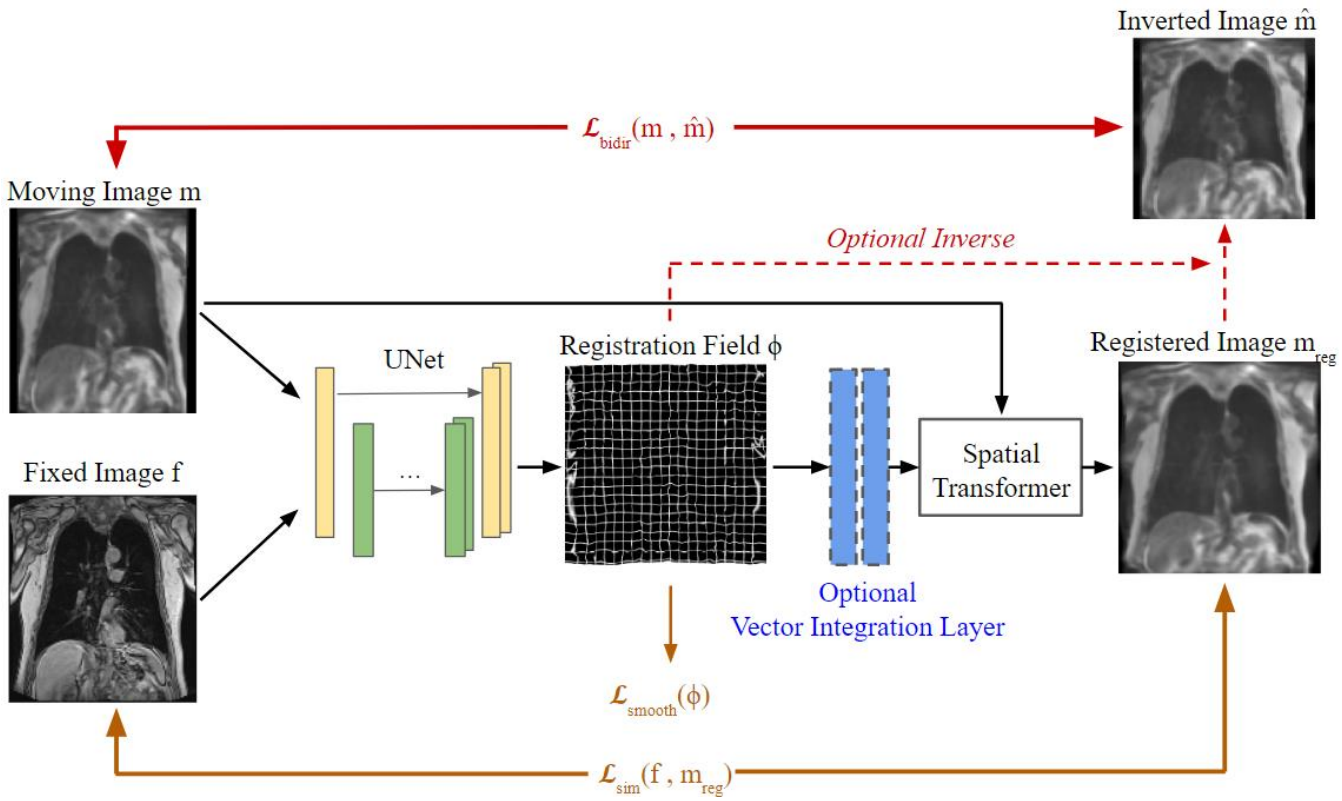
- U-Net structure
- 2 Loss Variants:
  - Unsupervised Loss
  - Supervised/Auxiliary Loss



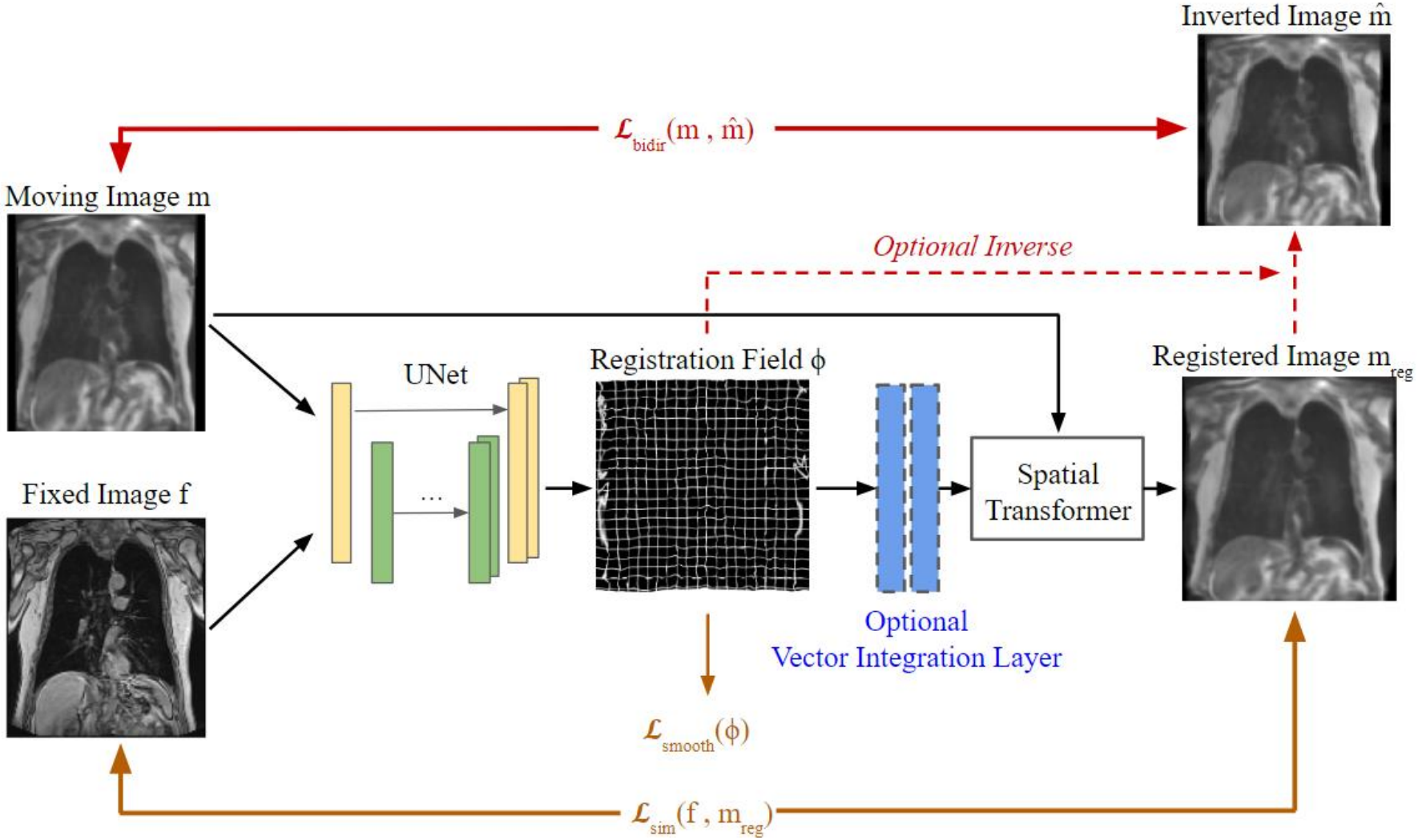
Guha Balakrishnan, 2019

Methods – VoxelMorph Implementation

- U-Net
- Spatial Transformer
- Loss Functions
- Vector Integration Layer



Methods – Overview



Methods – U-Net

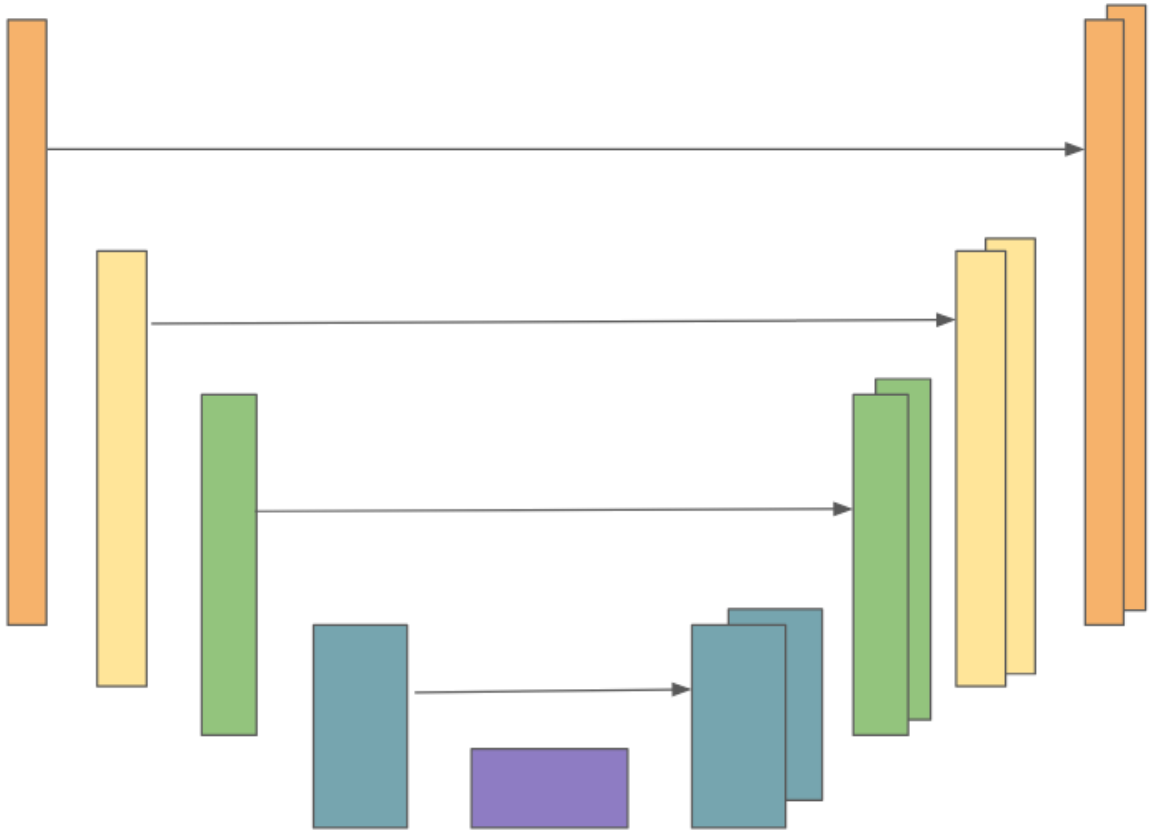
- Encoder

  Extract important hierarchical features while reducing spatial dimensions

- Decoder

  Expansion path to reconstruct dense, low-resolution feature maps back to the original resolution.

- Skip Connections



Methods – Encoder Downsample block

- 3D Convolution (Conv3D)
  - Kernel Size: 3x3, Stride: 2
  - Produces a feature map with increased channels for learned features
- Activation Function
  - Type: LeakyReLU , Parameter: 0.2
  - Adds non-linearity to the network to enhance learning capabilities
- Max Pooling Layer
  - Size: 2x2, Stride: 2
  - Reduces spatial dimensions by half, producing a coarse image for the next convolution layer

Methods – Decoder Upsample Block

- Up Convolution
  - Increases spatial resolution, learning more features during the reconstruction process
- Activation Layer
- Upsampling Layer
  - Doubles the spatial resolution
- Concatenation Layer
  - **Skip connections** combines features from the corresponding encoder layer
  - Increases the number of feature channels, enhancing the network's ability to reconstruct local details
  - Integrates local features from earlier layers with abstract features from deeper layers

Methods – Spatial Transformer

- Differentiable module with spatial transformation capabilities
- Calculates final voxel value base on the registration field
- **Linear Interpolation**
  - Estimate an integer final voxel value base on the neighboring voxels

$$m_{\text{reg}}(p) = \sum_{q \in Z(p')} m(q) \prod_{d \in \{x,y,z\}} (1 - |p'_d - q_d|)$$

- **Backpropagation**:
  - Interpolation process is differentiable, enabling gradient calculations for parameter optimization

Methods – Loss Functions

- Optimizes using only the input image pair and predicted registration field.
- No Labeled Data: operates without segmentation labels, making it fully unsupervised.

$$\mathcal{L}_{\text{us}}(f, m, \phi) = \mathcal{L}_{\text{sim}}(f, m_{\text{reg}}) + \lambda \mathcal{L}_{\text{smooth}}(\phi)$$

- Mean Squared Error

$$MSE(f, m_{\text{reg}}) = \frac{1}{|\Omega|} \sum_{p \in \Omega} [f(p) - m_{\text{reg}}(p)]^2$$

- Normalized Cross Correlation

$$NCC(f, m_{\text{reg}}) = \frac{\left(\sum_{p_i} \left(f(p_i) - \hat{f}(p)\right) \left(m_{\text{reg}}(p_i) - \hat{m}_{\text{reg}}(p)\right)\right)^2}{\left(\sum_{p_i} \left(f(p_i) - \hat{f}(p)\right)^2\right) \left(\sum_{p_i} \left(m_{\text{reg}}(p_i) - \hat{m}_{\text{reg}}(p)\right)^2\right)}$$

- Mutual Information

$$MI(f, m_{\text{reg}}) = \sum_{f,m} P_{f,m_{\text{reg}}}(f(p), m(p)) \log \left(\frac{P_{f,m_{\text{reg}}}(f(p), m(p))}{P_f(f(p)) \cdot P_{m_{\text{reg}}}(m(p))} \right)$$

- Bidirectional Loss

$$\mathcal{L}_{us}(f, m, \phi) = \mathcal{L}_{sim}(f, m_{\text{reg}}) + \omega \mathcal{L}_{bidir}(m, \hat{m}) + \lambda \mathcal{L}_{smooth}(\phi)$$

- Regularization Loss

$$L_{\text{smooth}}(\phi) = \sum_{p \in \Omega} \|\nabla u(p)\|^2$$

Methods – Vector Integration Layer

- Achieve a topology-preserving registration, diffeomorphism is required
- Defined by smooth and invertible transformations
- Vector Integration Layer converts the registration field into a diffeomorphic one
  - Treats the original registration field as a velocity field
  - Calculates diffeomorphism using *scaling and squaring* method

Methods – Experimental Setup

- 300 pair of morphological (fixed) and perfusion (moving) MRI
- Image size of 64 x 128 x 128
- 210 – 45 – 45 split
- Learning rate: 10^{-4}
- Batch size: 8
- Iterations: 1000

- **Validation Metrics**
  - Structural Similarity Test (SSIM)
  - Normalized Mutual Information (NMI)
  - Jacobian Test
  - Paired T-Test

- **Baseline Methods**
  - ANTs - Symmetric Normalization
  - Elastix - B-Spline Displacement

Methods – Experimental Setup

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- **Baseline Methods**
  - ANTs - Symmetric Normalization
  - Elastix - B-Spline Displacement
- **Accuracy**
  - Compare between fixed to registered
- **Invertibility**
  - Compare between moving to inverted

Results

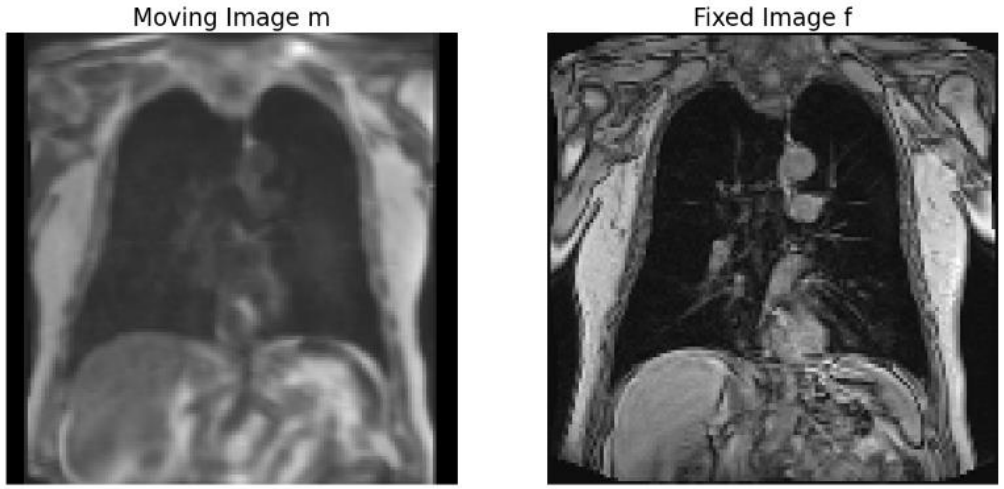
Experimental Variables

- 3 Loss Functions:
  - MSE, NCC, MI

- No Modification (NoMod)
- Vector Integration Layer (VecInt)
- Bidirectional Loss (Bidir)
- Both (VecBi)

- Regularization weight & placement

Reference Input Pair



NMI: 1.0903
SSIM: 0.3166

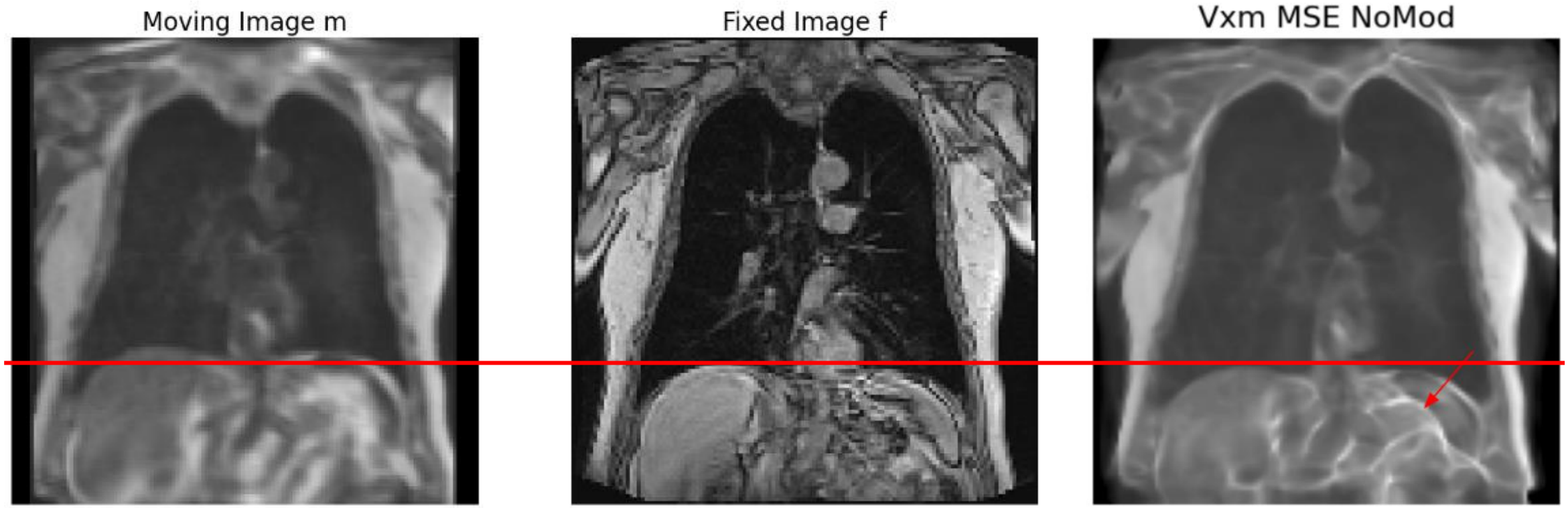
Results – MSE

Table 4.1.: Similarity metrics of MSE Setups

Setup	$\mathbf{NMI_{reg}}$	$\mathbf{NMI_{inv}}$	$\mathbf{SSIM_{reg}}$	$\mathbf{SSIM_{inv}}$	$\mathbf{det(J_\phi) \leq 0}$	$\mathbf{\%det(J_\phi) \leq 0}$
NoMod	**1.1444**	1.2743	**0.5931**	0.7685	9918	0.95
VecInt	1.1232	**1.4421**	0.5268	**0.9547**	**2188**	**0.21**
Bidir	1.1284	1.3734	0.5497	0.9130	5419	0.52

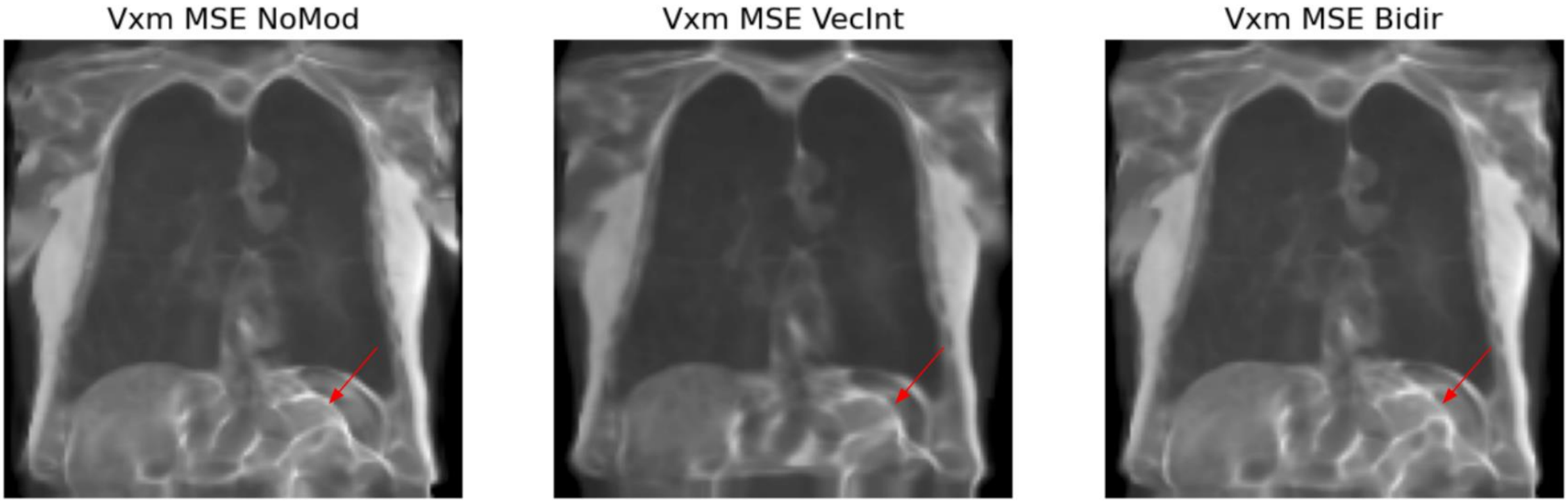
Results – MSE

- Undesirable artifacts



Results – MSE

- Undesirable artifacts



Results – NCC

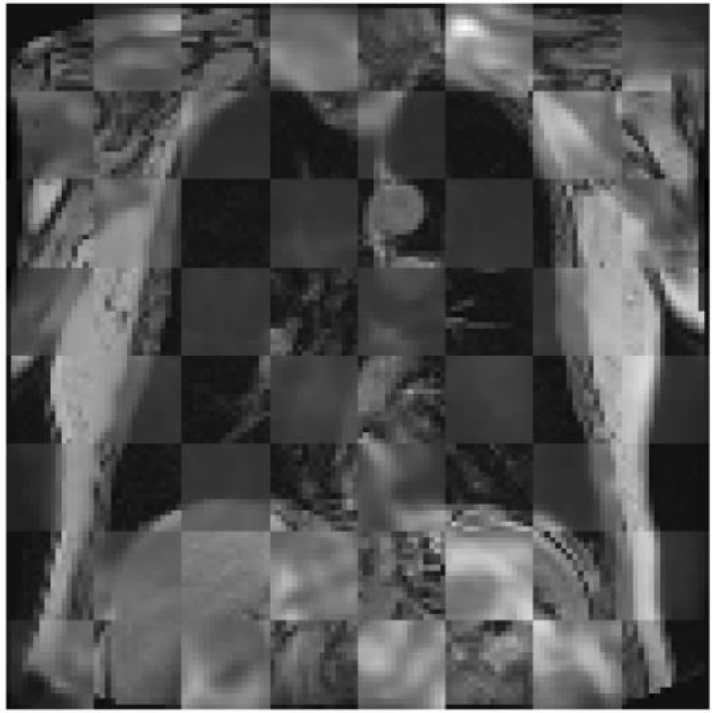
- Inverse Correlations of Accuracy to Invertibility
- Perfect Diffeomorphism

Table 4.2.: Similarity metrics of NCC Setups

Setup	$\mathbf{NMI_{reg}}$	$\mathbf{NMI_{inv}}$	$\mathbf{SSIM_{reg}}$	$\mathbf{SSIM_{inv}}$	$\mathbf{det(J_\phi) \leq 0}$	$\mathbf{\%det(J_\phi) \leq 0}$
NoMod	**1.1491**	1.2788	**0.6182**	0.7664	744	0.07
VecInt	1.1297	1.4413	0.5619	0.9218	**0**	**0**
Bidir	1.1465	1.2914	0.6138	0.7824	586	0.06
VecBi	1.1282	**1.4649**	0.5672	**0.9485**	**0**	**0**

Results – NCC

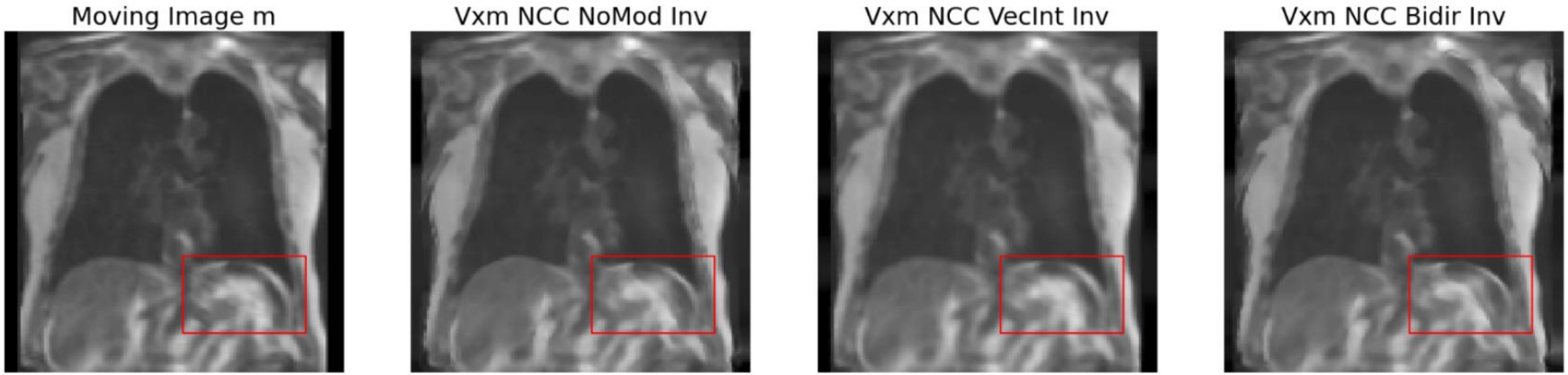
Vxm NCC NoMod



Vxm NCC VecInt

Vxm NCC Bidir

Results – NCC


Moving Image m
Vxm NCC NoMod Inv
Vxm NCC VecInt Inv
Vxm NCC Bidir Inv

Results – NCC

- Marginal Improvement with regularization parameters

Table 4.3.: Similarity metrics for different regularization locations in NCC VecInt models and their respective p-values and percentage increase(D-sample: Downsampled)

Method	NMI_{reg}	NMI_{inv}	$SSIM_{reg}$	$SSIM_{inv}$	$det(J_\phi) \leq 0$	$\%det(J_\phi) \leq 0$
Raw	1.1298	**1.4413**	0.5619	**0.9218**	0	0.00
D-sample	**1.1307**	1.4388	**0.5653**	0.9177	0	0.00
% inc.	0.08	-0.17	0.61	-0.44		
p-values	<0.001	0.0585	<0.001	<0.001		

Table 4.4.: Similarity Metrics of NCC VecInt with different regularization weight λ.

Method	NMI_{reg}	NMI_{inv}	$SSIM_{reg}$	$SSIM_{inv}$	$det(J_\phi) \leq 0$	$\%det(J_\phi) \leq 0$
$\lambda = 2.0$	1.1307	**1.4388**	0.5653	**0.9177**	**0**	0.00
$\lambda = 1.0$	**1.1388**	1.4202	**0.5824**	0.9028	2	0.00
% inc.	0.72	-1.29	3.02	-1.62		
p-values	<0.001	<0.001	<0.001	<0.001		

Results – MI

- Similar results to NCC

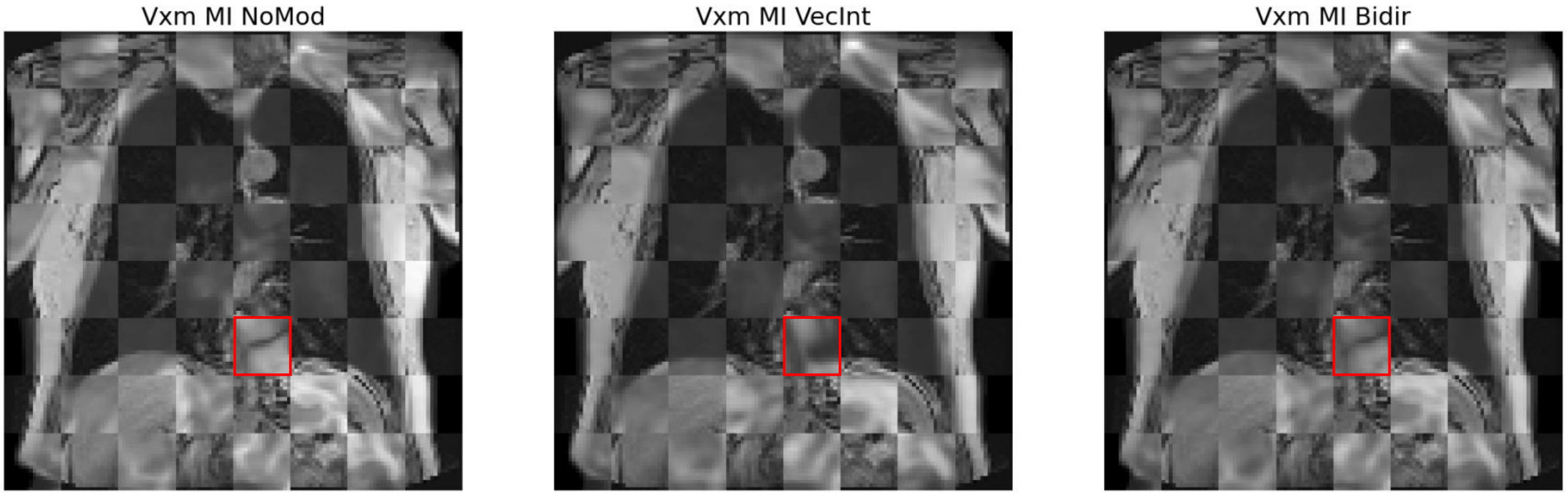
Table 4.5.: Similarity Metrics of MI Setups.

Method	NMI_{reg}	NMI_{inv}	$SSIM_{reg}$	$SSIM_{inv}$	$det(J_\phi) \leq 0$	$\%det(J_\phi) \leq 0$
NoMod	**1.1595**	1.3077	**0.5785**	0.8375	6715	0.64
VecInt	1.1420	1.4724	0.5257	0.9657	724	0.07
Bidir	1.1440	1.4178	0.5501	0.9458	4301	0.41
VecBi	1.1397	**1.5095**	0.5299	**0.9818**	**320**	**0.03**

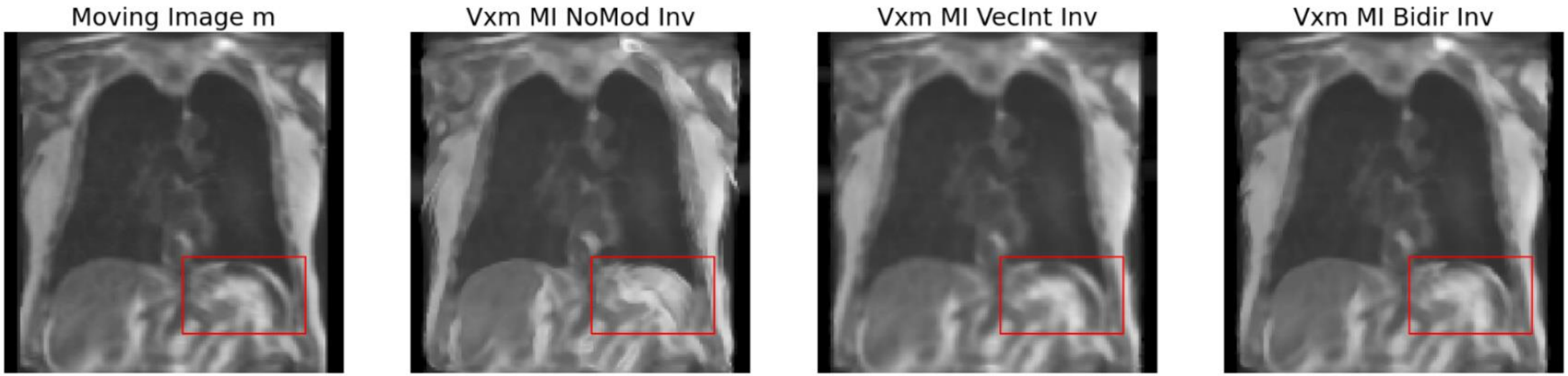
Table 4.6.: Similarity Metrics for different regularization locations in MI VecInt models and their respective p-values and percentage increase (D-Sample = Downsampled).

Method	NMI_{reg}	NMI_{inv}	$SSIM_{reg}$	$SSIM_{inv}$	$det(J_\phi) \leq 0$	$\%det(J_\phi) \leq 0$
Raw	1.1420	1.4724	0.5257	0.9657	724	0.07
D-Sample	1.1436	1.4659	0.5278	0.9640	816	0.08
% inc.	0.14	-0.44	0.40	-0.17		
p-values	<0.001	<0.001	0.0638	<0.01		

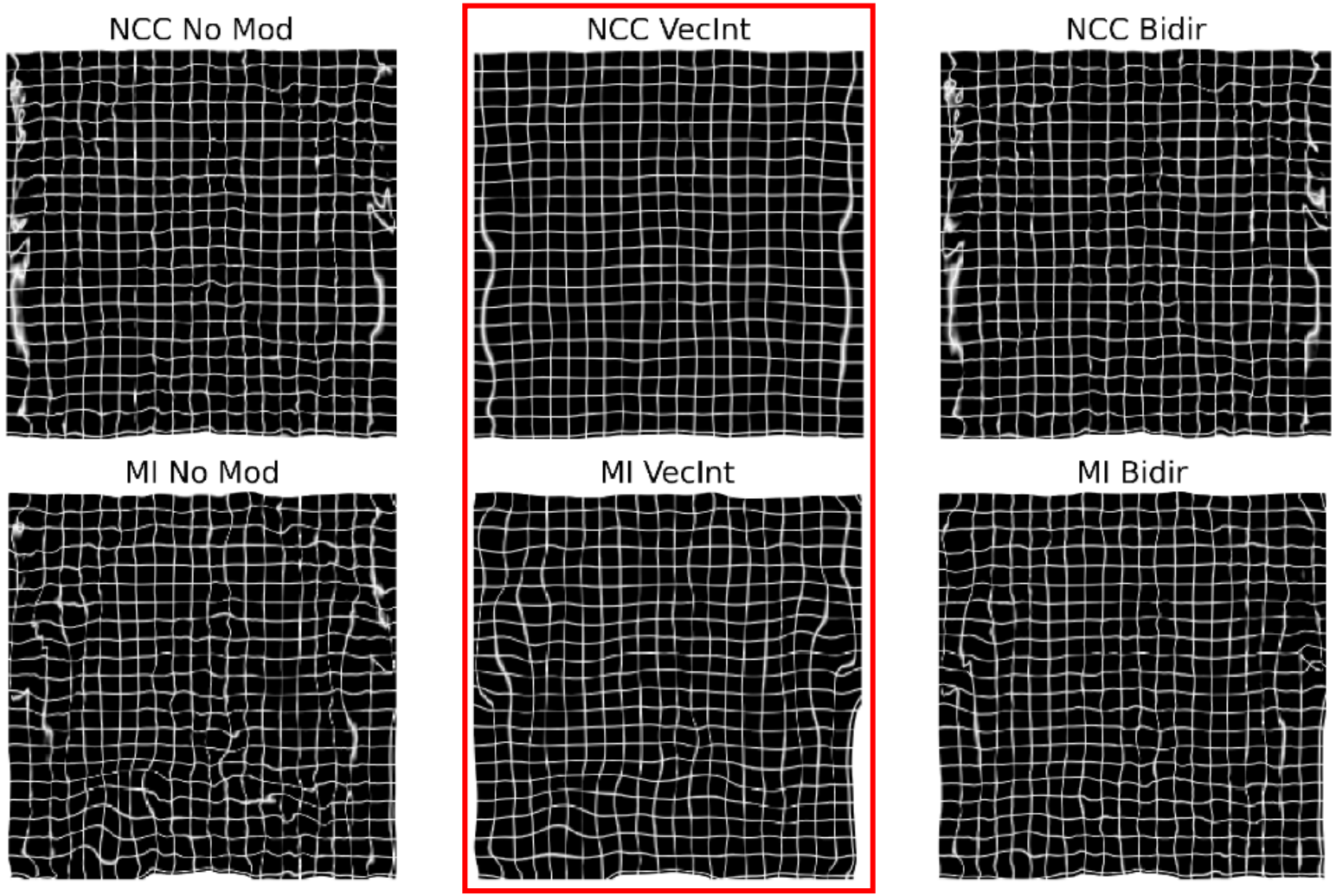
Results – MI



Results – MI


Moving Image m
Vxm MI NoMod Inv
Vxm MI VecInt Inv
Vxm MI Bidir Inv

Results – Registration Field Analysis



Results – Baseline Comparison

- Best performing models:
  - VXM_NCC_VecInt
  - VXM_MI_VecInt

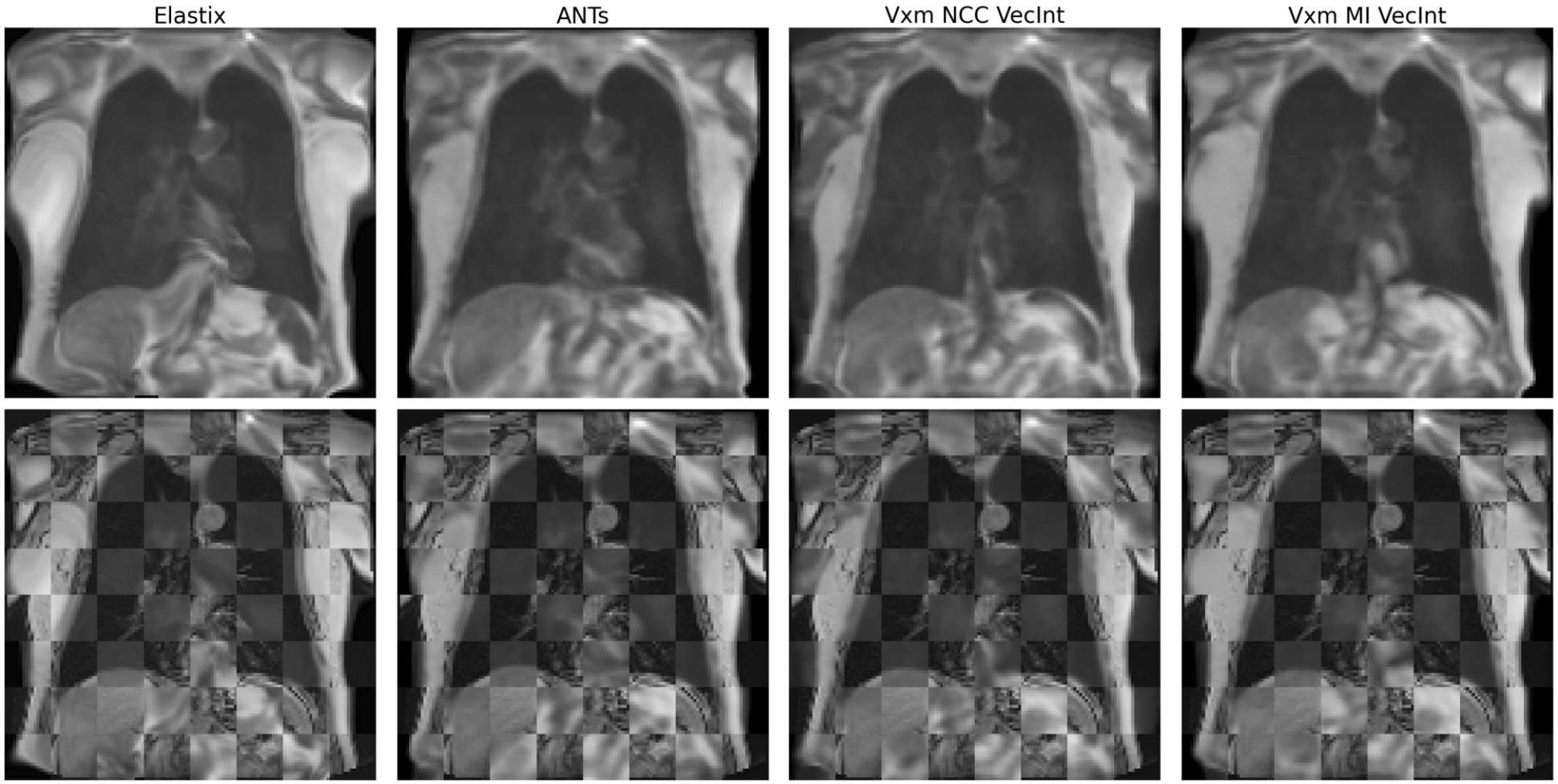
Table 4.8.: Registration Duration. DL:Deep Learning Methods.

Method	Time per sample	Time for whole sample
Elastix	350s	15,750s
ANTs	20s	900s
DL on CPU	2s	95s
DL on GPU	0.**38s**	**17s**

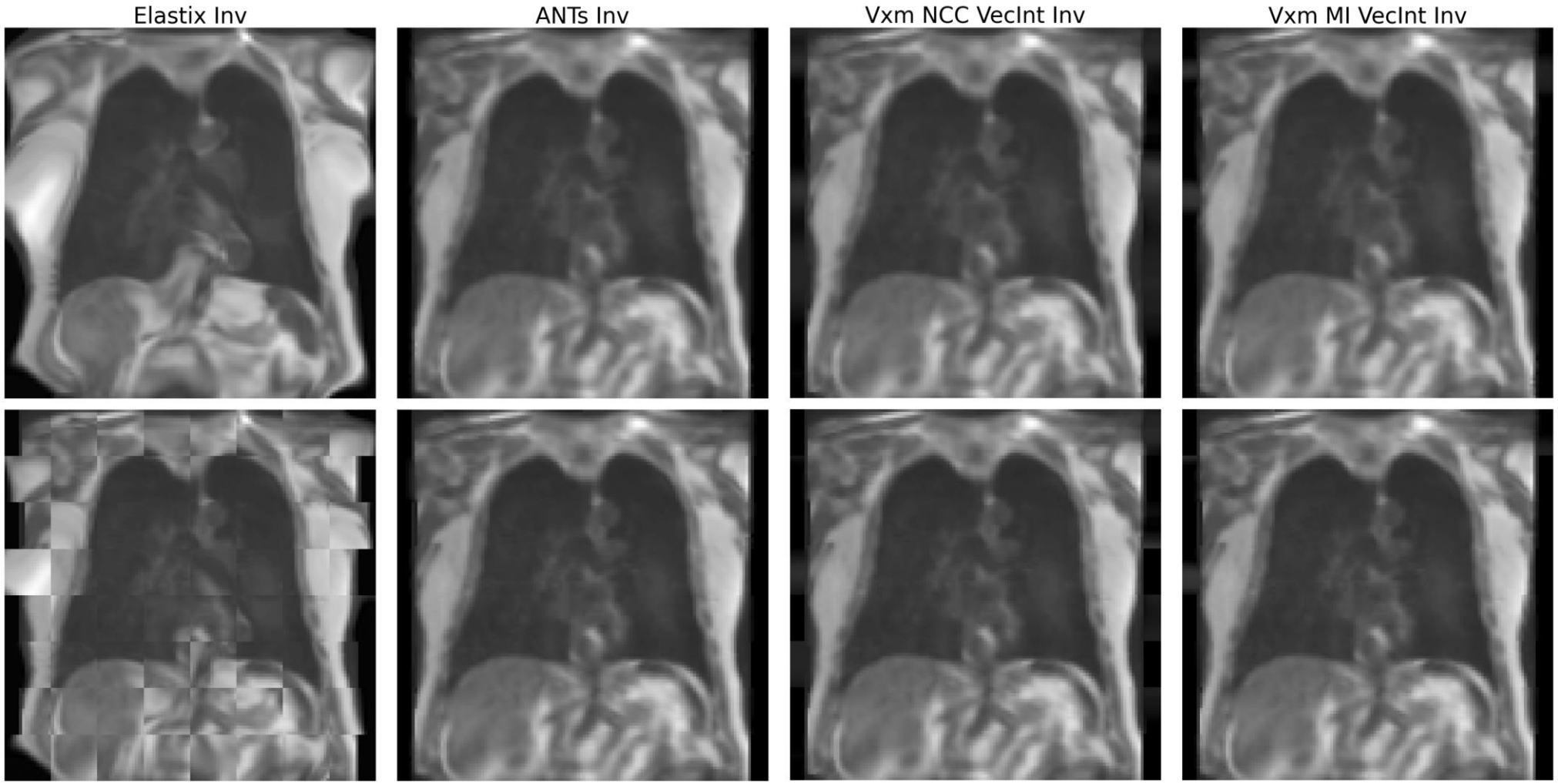
Table 4.7.: Similarity Metrics of Baseline Testing.

Method	NMI_{reg}	NMI_{inv}	$SSIM_{reg}$	$SSIM_{inv}$	$\det(J_\phi) \leq 0$	$\%\det(J_\phi) \leq 0$
Elastix	**1.1717**	1.0885	0.5721	0.4221	54373	5.19
ANTs	1.1379	**1.5063**	0.5321	**0.9760**	286	0.03
NCC	1.1418	1.4136	**0.5927**	0.8995	**2.7**	**0**
MI	1.1436	1.4711	0.5295	0.9648	589	0.06

Results – Baseline Registration Comparison

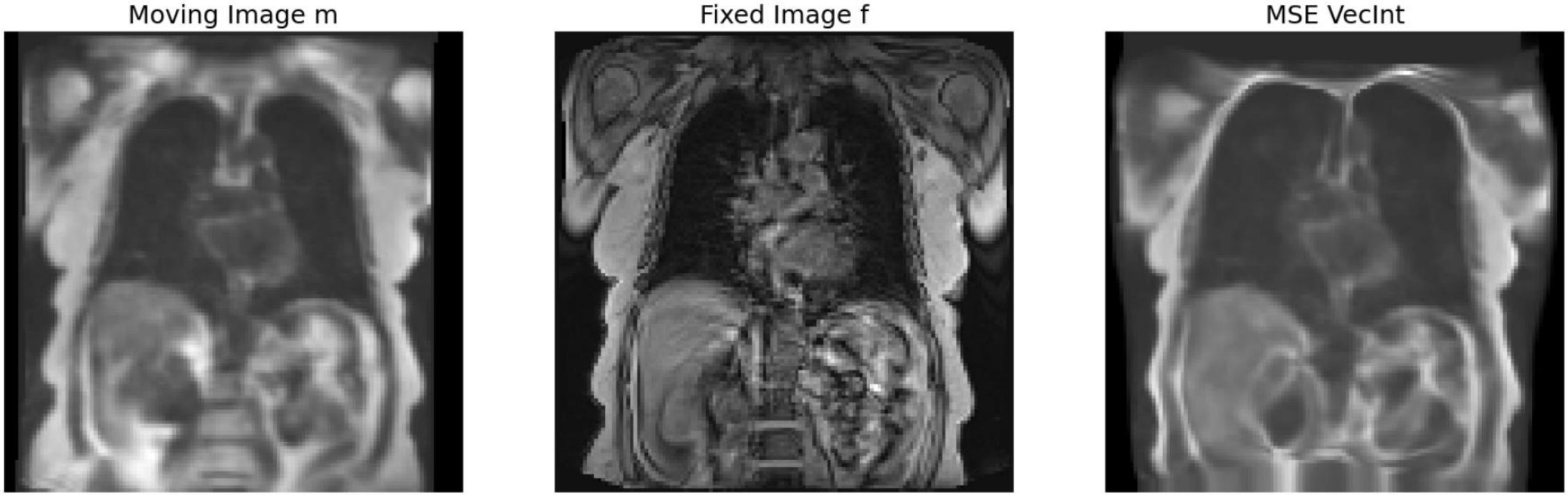


Results – Baseline Inversion Comparison



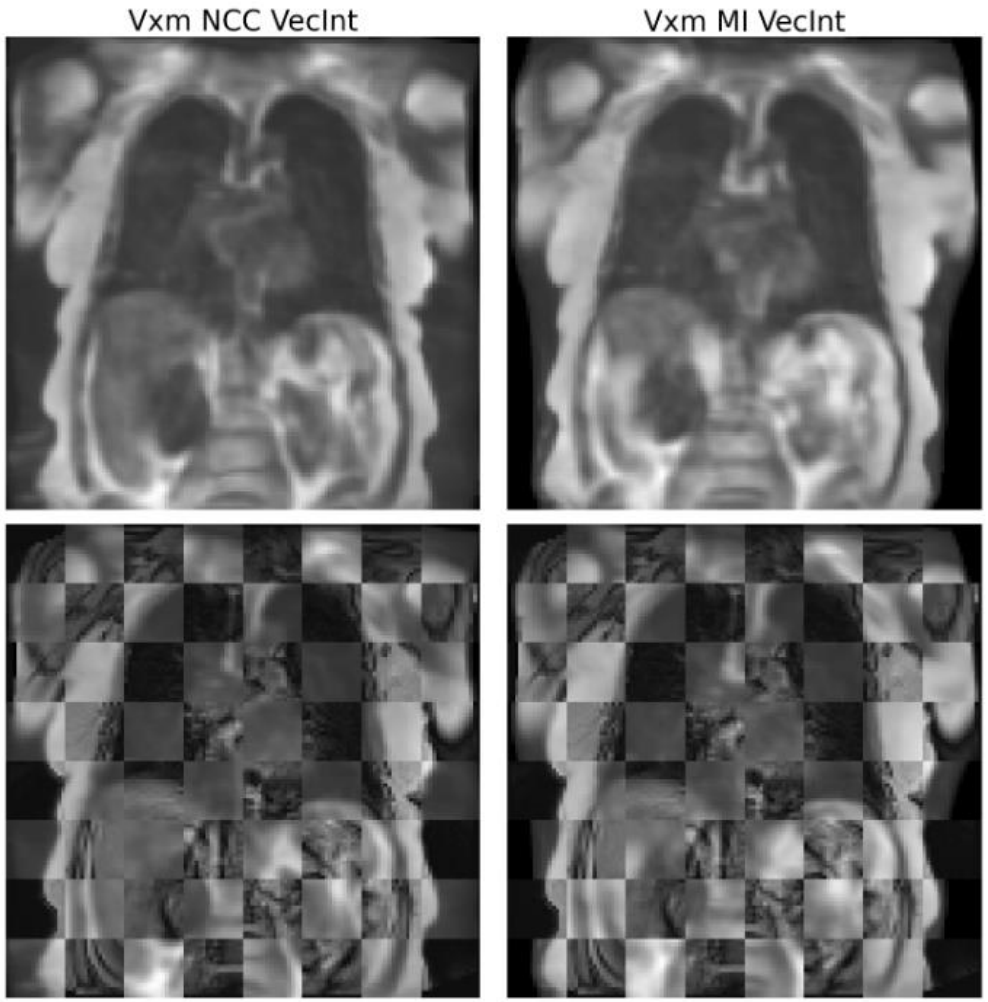
Discussion – Image Intensity

- Multi Modal Registration
- Undesirable artifacts
- Pure intensity-based loss function (MSE)



Discussion – Image Intensity

- NCC and MI performs without artifacts



Discussion – Bidirectional Weight Ratio

- Un-optimized
- MI (1 : 0.5), NCC (1:0.1)

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VecBi	1.1282	**1.4649**	0.5672	**0.9485**	**0**	**0**

Discussion – VecBi Potentials

- Proven to increase both accuracy and invertibility
- Currently un-optimized

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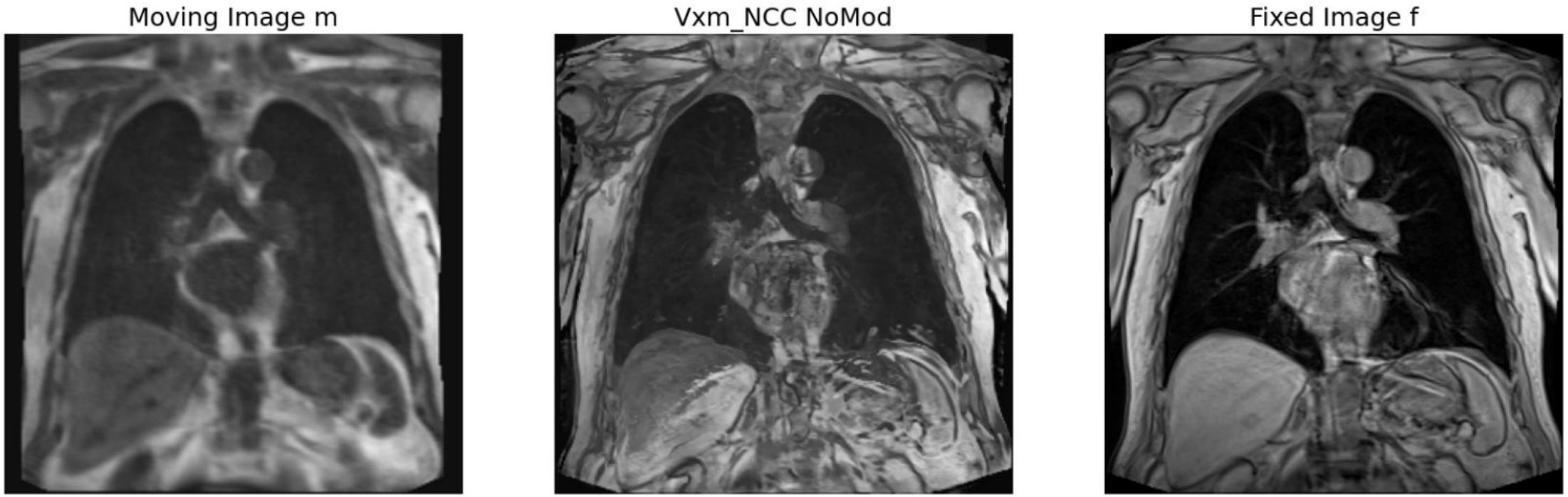
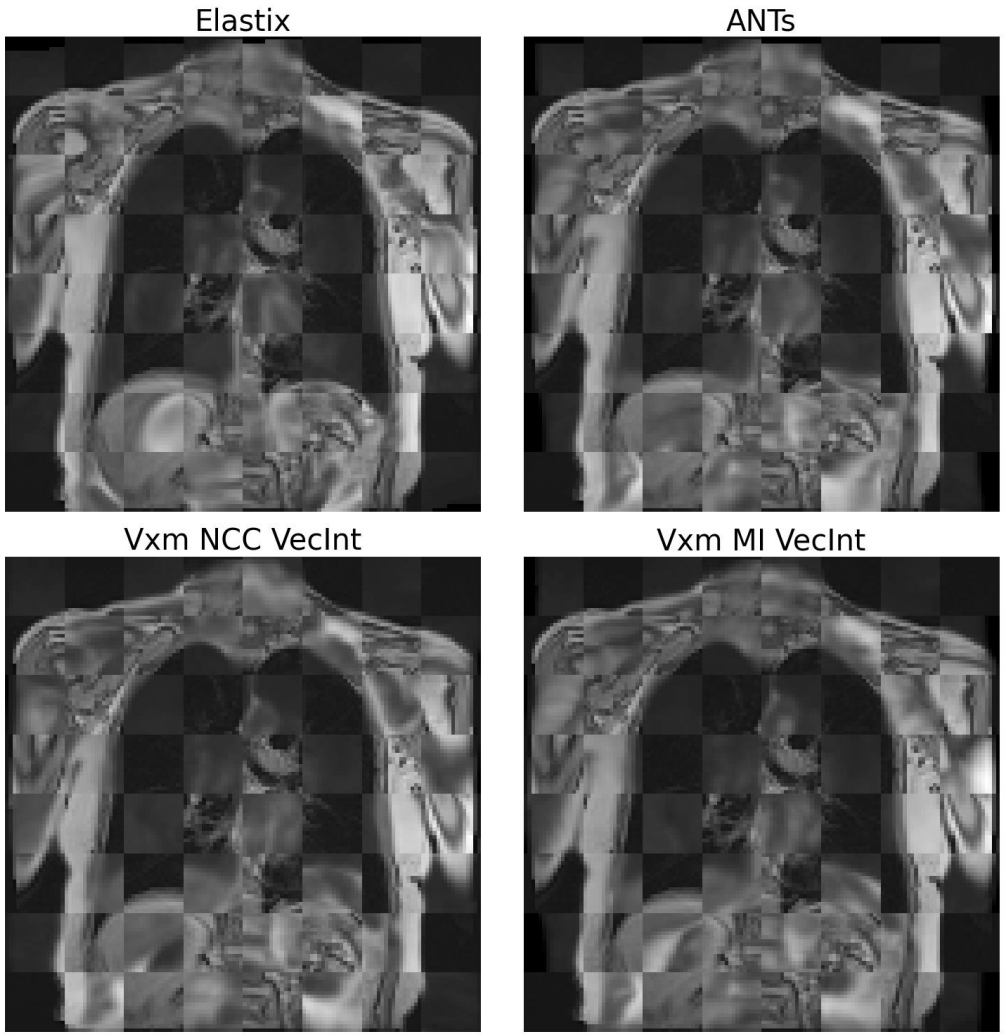
Discussion – Regularization Parameter

- Regularization parameter of 0.02
- Non-smooth registration
- No topology preservation



Discussion – CNN Receptive Field Limitations

- Unable to capture large deformations
- CNN kernel size limitations



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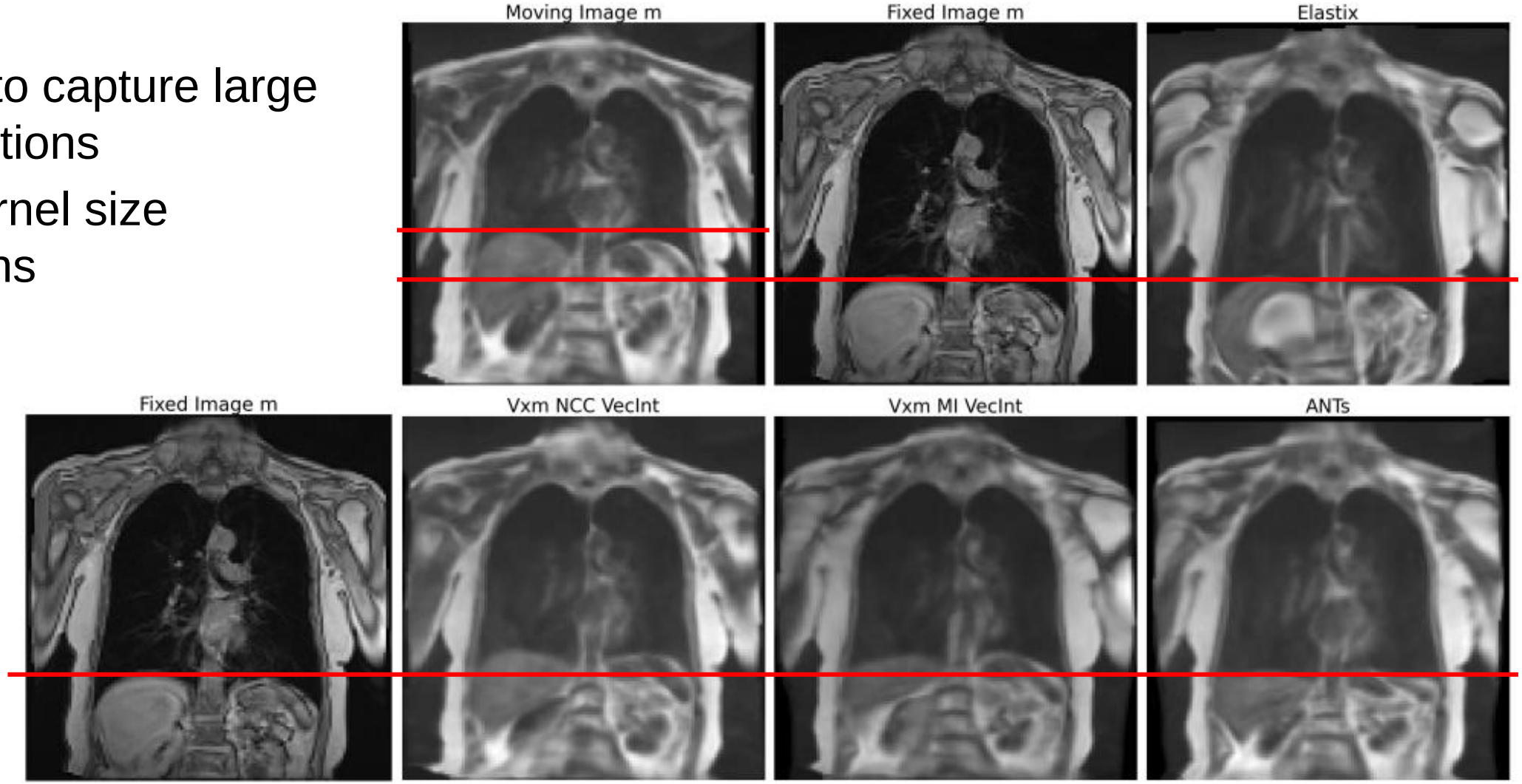


Conclusion

- Achieved similar results to VoxelMorph paper
- VoxelMorph is compatible with lung MRIs
- Comparable performance to ANTs and Elastix
- Significant Increase in registration speed

- More testing could be done
- More optimization could be done
- Implementation of DICE and auxiliary loss
- Testing out different deep learning models

Thank You!

- Any Questions?

State of Research

- **MRI- Magnetic Resonance Imaging**
  - Interaction between a magnetic field, radio waves, and Hydrogen atoms
  - B_0 magnetic field cause precession of Hydrogen atoms
  - Applying a second radiofrequency at the same frequency as the precession
  - Tipping the Hydrogen atoms out of alignment
  - Hydrogen atoms undergoes relaxation and emits characteristic frequency
  - Measured and converted into MRI

State of Research – Traditional Methods

- **Advanced Normalization Tools (ANTs)**:
  - Widely used toolkit for medical image registration
  - Affine (linear registration)
  - B-Spline Displacement Field (elastic registration)
  - Symmetric Normalization (SyN) (diffeomorphic registration)
  - Similarity measures: neighborhood cross-correlation, mean squared difference, and mutual information.
- **Elastix Toolbox**:
  - Commonly used method
  - Modular design
  - Does not support diffeomorphic registration

State of Research - Convolutional Neural Network

- Convolutional layers extract feature maps with trainable kernels.
- Activation layers apply non-linear functions
- Max pooling layers reduce spatial dimensions
- Dropout layers randomly set neurons to zero during training to prevent overfitting.
- Batch normalization normalizes inputs across layers for stable training.
- Multiple layers allow extraction of progressively abstract features.

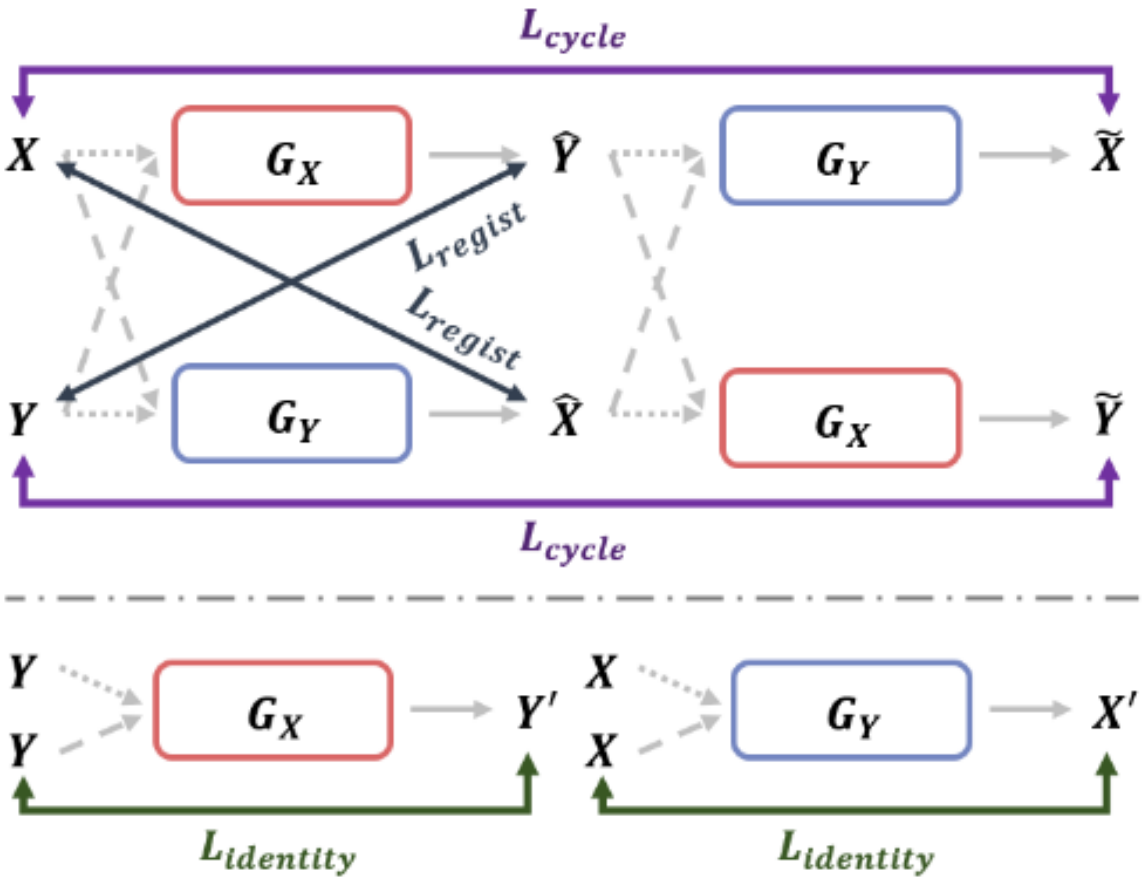
State of Research - CycleMorph

- Cycle Consistent structure

$$\mathcal{L}_{\text{regist}}(f, m, G_m) = \mathcal{L}_{\text{sim}}(f, m_{\text{reg}}) + \lambda \mathcal{L}_{\text{smooth}}(G_m)$$

$$\mathcal{L}_{\text{cycle}}(m, f, G_m, G_f) = \mathcal{L}_{\text{sim}}(\tilde{m}, m) + \mathcal{L}_{\text{sim}}(\tilde{f}, f)$$

$$\mathcal{L}_{\text{identity}}(m, f, G_m, G_f) = \mathcal{L}_{\text{sim}}(m', m) + \mathcal{L}_{\text{sim}}(f', f)$$



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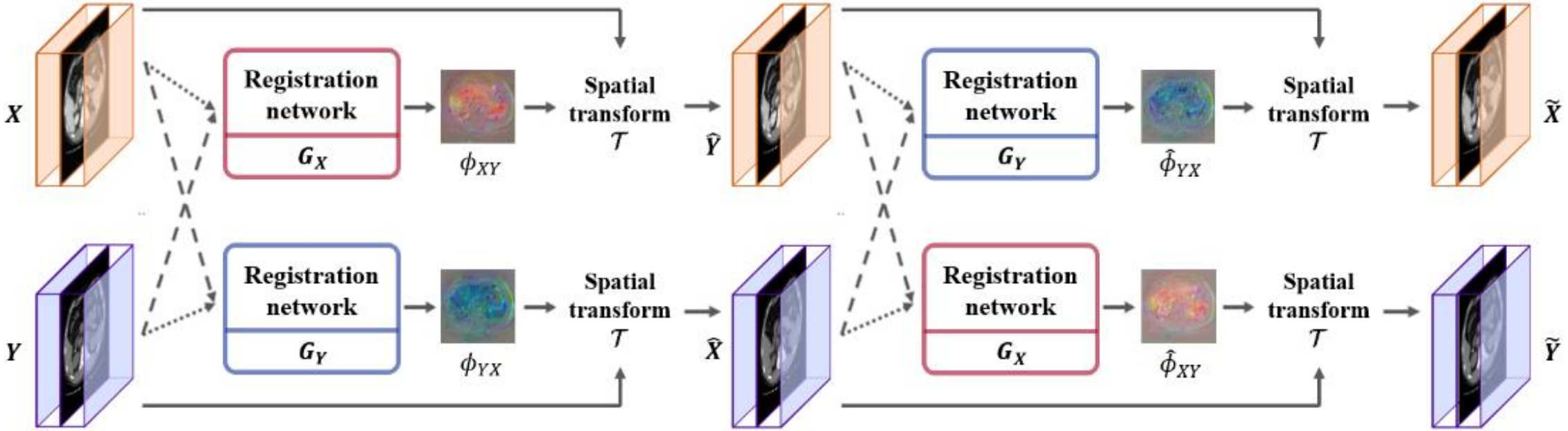
State of Research

- Parallel Network structure



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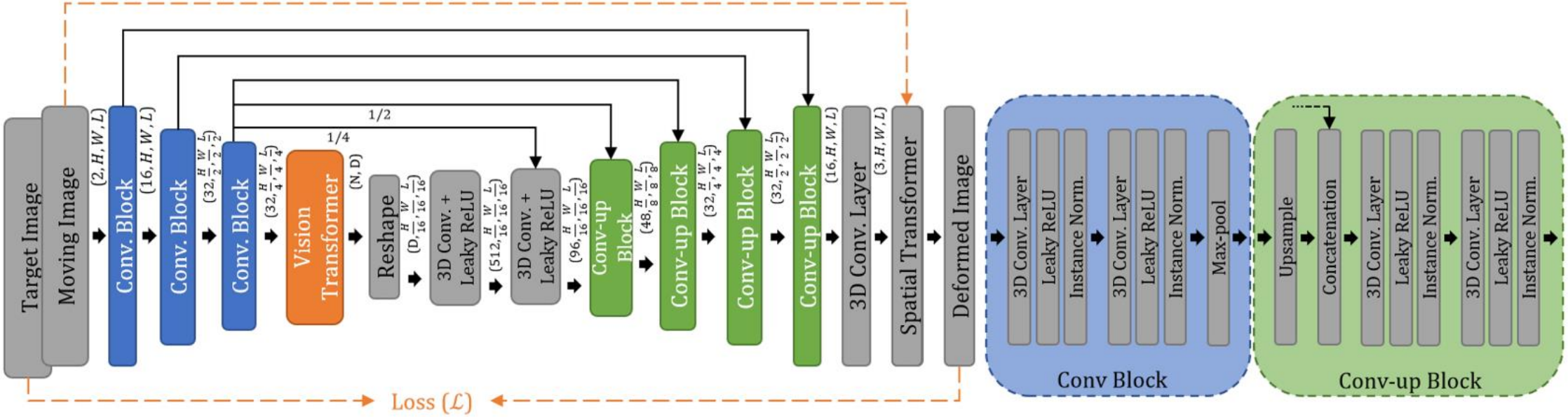
State of Research

Vit-V-Net



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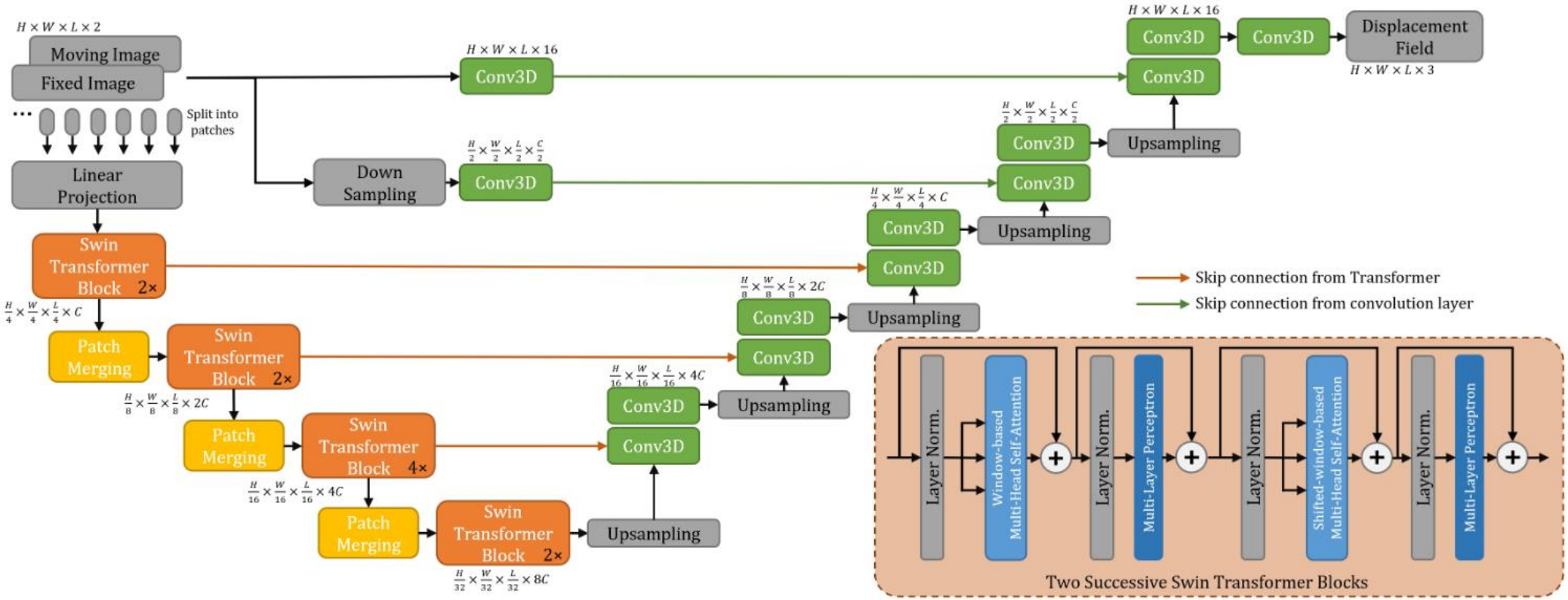
State of Research

TransMorph



Junyu Chen, 2022

Methods - Vector Integration Layer

- Stationary Velocity Field representation

$$\frac{\partial \phi^{(t)}}{\partial t} = u(\phi^{(t)})$$

- 2 Conditions:
  - One parameter sub-group of diffeomorphism
  - Lie algebra & Lie group

$$\phi^{(1)} = exp(u)$$

Methods - Vector Integration Layer

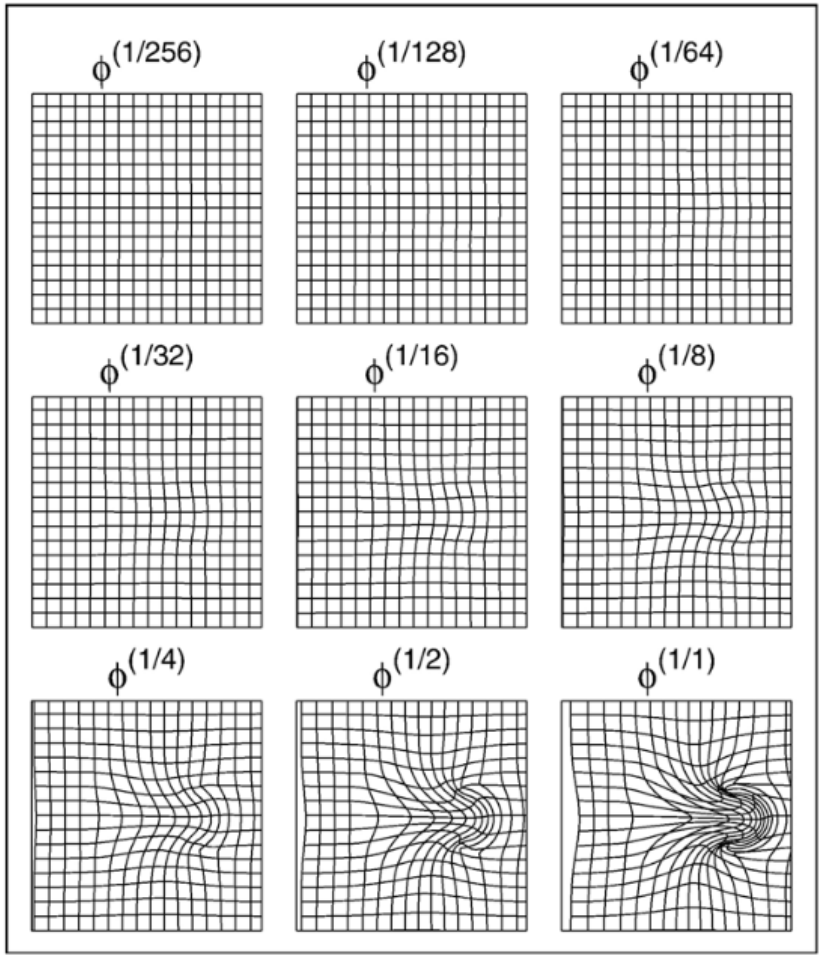
- **Scaling and Squaring**

1. $exp((t+t')u) = exp(tu) \circ exp(t'u)$

2. $exp(\frac{1}{2^{t-1}}u) = exp(\frac{1}{2^t}u) \circ exp(\frac{1}{2^t}u)$

3. $\phi_u^{(\frac{1}{2^{t-1}})} = \phi_u^{(\frac{1}{2^t})} \circ \phi_u^{(\frac{1}{2^t})}$

4. $\phi_u^{(1)} = \phi_u^{(\frac{1}{2})} \circ \phi_u^{(\frac{1}{2})}$



Methods – Validation Metrics

- **Structural Similarity Test**

$$SSIM(f, m_{\text{reg}}) = [l(f, m_{\text{reg}})^{\alpha}] \cdot \left[c(f, m_{\text{reg}})^{\beta}\right] \cdot [s(f, m_{\text{reg}})^{\gamma}]$$

- **Normalized Mutual Information**

$$NMI(f, m_{\text{reg}}) = \frac{H(f) + H(m_{\text{reg}})}{H(f, m_{\text{reg}})}$$

$$H(X) = -\sum_{x \in X} p(x) \log p(x)$$

- **Jacobian Test**

$$J_{\phi}(p) = \nabla \phi(p) \in R^{3 \times 3}$$